

python_object

Python Objects

Mutable

1. list
2. dictionary
3. set

Immutable

1. numeric
2. tuple
3. string
4. frozenset
5. Boolean

Numeric:-----

Integer:---

```
my_int1 = 10
my_int2 = 10
print(id(my_int1),id(my_int2))
```

| |

|

(140707225601224 140707225601224)

Same memory address

|

That means immutable object

Float:----

```
my_float1 = 10.5
my_float2 = 10.5
print(id(my_float1),id(my_float2))
```

| |

|

(1740399292944 1740399292944)

(Same memory address)

|

Immutable object

Complex:---

```
my_comp1 = 10.5+3j
my_comp2 = 10.5+3j
print(id(my_comp1),id(my_comp2))
```

```
| |
| |
-----
|
(3039707779024 3039707779024)
(Same memory address)
|
Immutable object
String:---
my_str1 = 'Neeraj'
my_str2 = 'Neeraj'
print(id(my_str1),id(my_str2))
| |
-----
|
Same memory address
(2421953832800 2421953832800)
|
That means immutable object
List:----
my_list1 = ['Neeraj','jai']
my_list2 = ['Neeraj','jai']
print(id(my_list1),id(my_list2))
| |
-----
|
(2070751895616 2070752043520)
Different memory address
|
That means mutable object
Tuple:---
my_tup1 = ('Neeraj','jai')
my_tup2 = ('Neeraj','jai')
print(id(my_tup1),id(my_tup2))
| |
-----
|
(1607757822976 1607757822976)
Same memory address
|
That means immutable object
Dictionary:---
my_dict1 = {'name':'Neeraj','age':37}
```

```
my_dict2 = {'name': 'Neeraj', 'age': 37}
print(id(my_dict1), id(my_dict2))
| |
-----
|
(2084796816704 2084797210368)
Different memory address
|
That means mutable object
Set: ---
my_set1 = {'name', 'Neeraj', 'age', 37}
my_set2 = {'name', 'Neeraj', 'age', 37}
print(id(my_set1), id(my_set2))
| |
-----
|
(2485072560864 2485072845888)
Different memory address
|
That means mutable object
FrozenSet: ---
my_fset1 = frozenset({'name', 'Neeraj', 'age', 37})
my_fset2 = frozenset({'name', 'Neeraj', 'age', 37})
print(id(my_fset1), id(my_fset2))
| |
-----
|
(2485072560864 2485072845888)
Different memory address due to unordered collection
|
Immutable object
Boolean: ----
my_bool1 = True
my_bool2 = True
print(id(my_bool1), id(my_bool2))
| |
-----
|
(140707224715696 140707224715696)
(Same memory address)
|
Immutable object
```